Durgakant Pushp

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Research Focus

Reliable autonomous vehicles and field robots operating under sensing, computational, and physical constraints. Research integrates visual navigation, localization, and bounded-rational decision-making with full-scale autonomous racing and robotic field deployment.

Education

Ph.D., Intelligent Systems Engineering **2021–2026**
Indiana University Bloomington

Dissertation: *Bounded Rational Robot: Embodiment, Representation and the Search for Intelligent Behavior*
Advisor: Prof. Lantao Liu

B.Tech., Computer Science and Engineering **2013–2017**
West Bengal University of Technology

Research and Professional Experience

Postdoctoral Research Fellow **2026–Present**
School of Mechanical Engineering, Purdue University; Purdue AI Racing (PAIR)

- Mentor a Ph.D. student on LiDAR-based localization for high-speed autonomous racing; guide research direction and technical development of the localization module.
- Contributed technical input across perception, planning, and control, and helped establish development goals and module priorities for PAIR's record-setting 2026 racing campaign.
- Contribute to rearchitecting the autonomy stack to support future research and expanded system capabilities; ongoing work.
- Develop visual autonomy for emergency stopping when other sensing modalities become unavailable, extending prior IJRR work on visual decision-making to racing. Simulation results obtained; research ongoing.
- Search over embodiment: A bounded rational agent should reason about its embodiment fitness for a particular task; this ongoing work aims to find a solution.

Team Lead, IU–Luddy Autonomous Racing Team **2023–2026**
Indiana University Bloomington

- Directed day-to-day technical development and project execution for a full-scale autonomous racing program, with substantial autonomy in research planning, task allocation, and integration.
- Led autonomy stack development spanning localization, perception, planning, control, and data visualization for the IAC AV-24 platform.
- Led and mentored a multidisciplinary team of doctoral, master's, undergraduate, and high-school students; coordinated development and on-track testing at IMS, LVMS, Laguna Seca, and Kentucky Motor Speedway.
- Team achieved 128 mph within one month of preparation, second place at the 2025 LVMS time trial with a 164 mph average, and fourth place at the 2025 Laguna Seca road-course event.
- Contributed proposal sections supporting the racing program and led development underlying the ISER autonomous racing stack publication.

Research and Professional Experience (continued)

Research Assistant; Project Technical Leadership **2023–2025**
Indiana University Bloomington; ERDC-supported robotic inspection of flood-control systems

- Led day-to-day research and project management for IU's work on robotic inspection of flood-control infrastructure, including research planning, student coordination, and field experimentation.
- Developed a UAV–miniUGV deployable system for confined-space inspection and designed and field-tested MiniUGV$_2$, integrating mobility, manipulation, sensing, and mission autonomy.
- Worked within a multi-university collaboration involving TAMU, Rice, UT Austin, LSU, MSU, and IU; contributed proposal sections for the project.

Research Assistant; Project Technical Leadership **2021–2023**
Indiana University Bloomington; ARL-supported Off-Road Autonomy / SARA Program

- Led day-to-day technical execution and managed research progress for IU's off-road autonomy work, including terrain perception and autonomous navigation.
- Developed autonomy modules transitioned into ARL's Robotics Enterprise software repositories; contributed proposal sections supporting the project.

Team Lead, AISUM Autonomous Systems Challenge **2021–2022**
Indiana University Bloomington; U.S. Navy CRANE / OUSD(R&E)

- Led IU's single-member entry on GPS-denied exploration to a national Top-8 finish; contributed proposal sections and challenge reports.

Research Engineer; Field Experiment Lead **2017–2020**
TCS Research & Indian Institute of Science, Bangalore, India

- Developed perception, navigation, aerial manipulation, and decentralized multi-robot coordination software for UAV/UGV systems.
- Led MBZIRC Challenge 1 work and field experiments, including extensive outdoor flight testing; team received the MBZIRC 2020 Best Innovation Award.

Research Mentoring and Scholarly Leadership

Selected contributions to student research and publication development.

- **LiDAR localization for high-speed racing, Purdue.** Mentor a Ph.D. student on the localization module and its research development.
- **Seeing Where to Deploy (IROS 2026).** Directed the research from project inception through manuscript completion; guided the first and second authors on research direction and execution, and contributed to writing.
- **AFRDA (RA-L 2025).** Held weekly research meetings, helped shape the study through submission, and contributed to manuscript writing.
- **From Zero to High-Speed Racing (ISER 2025).** Led the underlying racing-system development and guided the work through most of the project; the first author led final completion for submission.

Publications

My name is in bold. Preprints are listed separately.

Journal Articles

[J1] **D. Pushp**, W. Chen, Z. Chen, C. Luo, J. M. Gregory, and L. Liu. "Navigating the Wild: Pareto-Optimal Visual Decision-Making in Image Space." *The International Journal of Robotics Research (IJRR)*, 2025.
Paper of the Season, Q2 2025: Honorable Mention.

[J2] M. A. M. Khan, **D. Pushp**, and L. Liu. "AFRDA: Attentive Feature Refinement for Domain Adaptive Semantic Segmentation." *IEEE Robotics and Automation Letters (RA-L)*, 2025.

[J3] Z. Chen, **D. Pushp**, J. M. Gregory, and L. Liu. "Pseudo-Trilateral Adversarial Training for Domain Adaptive Traversability Prediction." *Autonomous Robots*, vol. 47, pp. 1155–1174, 2023.

Conference and Symposium Publications

[C1] H. Jardali, I. S. Mohamed, **D. Pushp**, and L. Liu. "LPV-MPC for Lateral Control in Full-Scale Autonomous Racing." *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[C2] S. Lee, S. M. Shithil, **D. Pushp**, L. Liu, and Z. Wang. "Seeing Where to Deploy: Metric RGB-Based Traversability Analysis for Aerial-to-Ground Hidden Space Inspection." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

[C3] **D. Pushp**, S. Kalhapure, S. M. Shithil, and L. Liu. "MiniUGV$_2$: A Compact UAV-Deployable Tracked Ground Vehicle with Manipulation Capabilities." *International Symposium on Experimental Robotics (ISER)*, 2026.

[C4] H. Jardali, **D. Pushp**, Y. Yu, M. Ali, I. S. Mohamed, A. Murillo-Gonzalez, P. D. Coen, M. A.-M. Khan, R. C. Pulivendula, S. Park, L. Zhou, and L. Liu. "From Zero to High-Speed Racing: An Autonomous Racing Stack." *International Symposium on Experimental Robotics (ISER)*, 2025.

[C5] **D. Pushp**, J. Xu, Z. Chen, and L. Liu. "Context-Generative Default Policy for Bounded Rational Agents." *IEEE/RSJ IROS*, 2024.

[C6] M. Ali, **D. Pushp**, Z. Chen, and L. Liu. "Visual-Geometry GP-Based Navigable Space for Autonomous Navigation." *IEEE/RSJ IROS*, 2024.

[C7] **D. Pushp**, Z. Chen, C. Luo, J. M. Gregory, and L. Liu. "POVNav: A Pareto-Optimal Mapless Visual Navigator." *ISER*, 2023.

[C8] **D. Pushp**, J. Xu, and L. Liu. "Coordination of Bounded Rational Drones through Informed Prior Policy." *IEEE/RSJ IROS*, 2023.

Publications (continued)

[C9] **D. Pushp**, S. Kalhapure, K. Das, and L. Liu. "UAV-miniUGV Hybrid System for Hidden Area Exploration and Manipulation." *IEEE/RSJ IROS*, pp. 1297–1304, 2022.

[C10] J. Xu, **D. Pushp**, K. Yin, and L. Liu. "Decision-Making Among Bounded Rational Agents." *International Symposium on Distributed Autonomous Robotic Systems (DARS)*, 2022.

[C11] Z. Chen, **D. Pushp**, and L. Liu. "CALI: Coarse-to-Fine Alignments for Unsupervised Domain Adaptation of Traversability Prediction." *Robotics: Science and Systems (RSS)*, 2022.

[C12] U. Rai, **D. Pushp**, A. Choudhary, A. Kumar, B. Chaitanya, R. Lima, N. S. Abhinay, R. Majumder, S. Jana, K. Das, and D. Ghose. "PARA-FM: Graphical User Interface for Emergency Flood Planning and Management." *Proceedings of the 5th World Congress on Disaster Management*, vol. II, pp. 100–106, Routledge, 2022.

[C13] R. Srivastava, R. Lima, C. Shinde, **D. Pushp**, and K. Das. "Estimation and Control for Autonomous UAV System to Neutralize Unknown Aerial Maneuvering Targets." *7th Indian Control Conference (ICC)*, pp. 117–122, 2021.

[C14] V. Anand, **D. Pushp**, R. Raj, and K. Das. "Gaussian Mixture Model (GMM)-Based Object Detection and Tracking Using Dynamic Patch Estimation." *IEEE/RSJ IROS*, pp. 4474–4481, 2019.

[C15] V. Anand, **D. Pushp**, R. Raj, and K. Das. "Gaussian Mixture Model (GMM)-Based Dynamic Object Detection and Tracking." *International Conference on Unmanned Aircraft Systems (ICUAS)*, pp. 1365–1371, 2019.

Preprint

M. Ali, H. Jardali, Y. Yu, **D. Pushp**, and L. Liu. "Minimalistic Autonomous Stack for High-Speed Time-Trial Racing." arXiv:2509.19636, 2025.

Other Conference Contribution

D. Pushp and S. Samaddar. "A Randomly Guided Algorithm for Exact Pattern Matching." National Conference on Recent Advances in Science and Technology (NCRAST), Heritage Institute of Technology, Kolkata, September 27–29, 2016.

Honors and Awards

2026 IJRR Paper of the Season Honorable Mention, Q2 2025.

2024 Second Place, Cheng Wu Innovation Challenge; single-member team.

2023 Audience Choice Award, MassRobotics Form & Function Challenge; single-member team.

2021–2022 Top-8 National Finalist, AISUM Challenge; single-member team.

2014, 2015, 2017 Boeing-sponsored RC aircraft competition awards, IIT Kharagpur.

2015, 2016 First Prize, RC Aircraft Flying Competition, NIT Jamshedpur.

2016 Globsyn Faculty Choice Award for analysis of aircraft crash data.

Team Competition Results

2026 PAIR: First Place, Time Trial; Second Place, Multi-car Race, Indy Autonomous Challenge, Laguna Seca; record-setting lap.

2025 IU–Luddy: Second Place, LVMS Time Trial; Fourth Place, Laguna Seca Road Course, Indy Autonomous Challenge.

2020 Best Innovation Award, Mohamed Bin Zayed International Robotics Challenge.

Teaching Experience

Guest Lecturer, Autonomous Robots **August 2024**
Intelligent Systems Engineering, Indiana University Bloomington

Undergraduate/graduate course; approximately 20 students.

- Delivered a lecture on autonomous racing, covering full-stack autonomy, GNSS-based localization, PID and Pure Pursuit control, and an introduction to MPPI.

Teaching Assistant, Introduction to Cyber-Physical Systems **January–May 2024**
Intelligent Systems Engineering, Indiana University Bloomington

Undergraduate course; approximately 30 students.

- Supervised student projects using ROS and autonomy stacks, held office hours, and supported assessment.

Workshop Organizer and Instructor, Aeromodelling **September 2017**
Jadavpur University, Kolkata, India

Two-day workshop; approximately 250 undergraduate participants.

- Designed and delivered lectures on aerodynamics and flight mechanics; guided hands-on construction and flight of RC aircraft models.

Patents

- R. Lima, **D. Pushp**, et al. "Robot-Mountable Adaptive Gripper Assembly for Intercepting Aerial Targets." Indian Patent No. IN 202321018297 (2024).
- **D. Pushp**, S. Kalhapure, and L. Liu. "UAV Tethered to UGV System." U.S. provisional patent application No. 63/914,830.

Professional Service and Outreach

- Peer reviewer: ICRA, IROS, IEEE Robotics and Automation Letters, and ISER.
- Research and robotics mentoring for graduate, undergraduate, and high-school students through autonomous systems projects.

Technical Expertise

Autonomy: localization; visual navigation; terrain perception; planning and control; multi-robot coordination; field testing and systems integration.
Programming and robotics: C++, Python; ROS 1/2, Gazebo, AirSim, MuJoCo.
Learning and vision: PyTorch, TensorFlow, OpenCV, scikit-learn.
Sensing: LiDAR, radar, RTK-GNSS, IMUs, wheel encoders, depth and localization cameras, event-based cameras, sonar, and proximity sensors.